

5.3 ServiceNow Testing

1. Testing Framework & Methodology

Testing for the SCMS application covers three distinct validation layers:

1. **Unit Testing:** Automated testing of server scripts (`SCMSUtils`), Business Rules, and Client Scripts.
2. **Integration & Flow Testing:** Verification of Flow Designer triggers, approvals, and dynamic email notifications.
3. **Security & Access Control Testing:** Automated and manual evaluation of Access Control Lists (ACLs) across custom scoped tables (`x_scms_*`).

2. Test Execution Breakdown

A. Automated Test Framework (ATF) & Server Scripting

- **Script Includes Verification (`SCMSUtils`):** Tested date-difference math functions and leave balance check logic with 100% code coverage.
- **Business Rule Validation:** Confirmed `before insert` script on `x_scms_enrollment` successfully halts duplicate records before database write.

B. Client Script & UI Policy Tests

- **Date Validation:** Form submission blocked with field-level notification when `End Date < Start Date`.
- **Dynamic Field Visibility:** Mandatory UI policies dynamically enforce mandatory reason fields when `Leave Type = Medical`.

C. Role & Security Access Control (ACL) Testing

- Executed test runs impersonating users with roles `x_scms.student`, `x_scms.faculty`, and `x_scms.staff` to ensure zero privilege escalation.